

SSPA Specialty - Calculation Questions Sprint for Full Score

T1 more number of digits + T2 decimal + T9 fraction • Three major calculation traps total attacks • 75 minutes • 1-to-3 Online Lesson

Corresponding textbooks: Calculation questions in the first test paper (accounting for about 40-50% of the paper)

• Multi-digit • Decimal • Fraction

Core traps: □ T1 Multi-digit calculation • T2 Decimal operation • T9 Fraction operation - ● SSPA compulsory exam

SSPA related: ● High frequency The average score loss rate for the first calculation question in the scored paper is as high as 35%, mostly due to carelessness rather than ignorance

Prerequisite knowledge: L1-L3 multi-digit • L12-L13 decimal • L7-L10 fraction

|||SEP||| ❶ Multiple digits + zero decimal errors ❷ Automation of common fractions ❸ Comprehensive calculation verification habits ❶ Zero mistakes in multi-digit + decimal places ❷ Automated fraction division ❸ Comprehensive calculation verification habits

Student Name:

Class:

Date:

Time Spent:

I. calculate warm-up (total 5 questions, 5 minutes)

#	Question	Difficulty	Working Space
W1	Calculate $12,345 + 987,654 = ?$		
W2	Calculate $3.4 + 0.56 = ?$ (Align the decimal points!)		
W3	Calculate $\text{LCM}(6, 8) = ?$		
W4	Calculate $0.6 \times 5 = ?$		
W5	Divide $\frac{3}{4}$ and $\frac{5}{6}$ (write two equivalent fractions).		

II. Core Knowledge + Worked Examples

Knowledge point 1: Multi-digit + decimal calculation traps (T1 + T2) SSPA required exam

The four iron laws of multi-digit calculations:

- ① **Alignment rules** |||SEP|||: **Addition and subtraction** → **Align the same digits (ones digits to ones, tenths to tenths)** Carry mark |||SEP|||: Mark "small 1" on the previous line when adding and carrying, and mark "small dot" when subtracting and borrowing
- ② **Decimal addition and subtraction** |||SEP|||: **The decimal points must be aligned! Fill in the missing digits with 0 (for example, $3.4 + 0.56 \rightarrow 3.40 + 0.56$)** Decimal multiplication |||SEP|||: Multiply integers first → Count the total decimal places → Point the decimal point from right to left
- ③ **Verification habits** |||SEP|||: **Mental arithmetic estimation (for example, $12,345+987,654 \approx 120,000 + 990,000 = \text{approximately } 1.11 \text{ million}$)**: The decimal points must be aligned! Fill in missing bits with 0 (e.g. $3.4 + 0.56 \rightarrow 3.40 + 0.56$)
- ④ **Decimal multiplication**: First multiply integers → count the total number of decimal places → click the decimal point from right to left
- ⑤ **Verification Habits**: Mental arithmetic estimation (for example, $12,345+987,654 \approx 120,000+990,000 = \text{about } 1.11 \text{ million}$)

Example KP1-1: Multi-digit addition (including carry)

Calculation: $2,345,678 + 8,765,432 = ?$

Common mistakes

10,110,110

Forgot to add to the previous level when carrying

Correct solution

11,111,110

Bit-by-bit alignment, carry mark at each level, level-by-level inspection

Example KP1-2: Decimal multiplication trap

Calculation: $0.35 \times 0.4 = ?$

❌ Common mistakes

$$0.35 \times 0.4 = 1.40$$

$35 \times 4 = 140$, just write 140 and forget the decimal point

✅ Correct solution

$$0.35 \times 0.4 = 0.140 = 0.14$$

$35 \times 4 = 140$, $2+1=3$ decimal places $\rightarrow 0.140 \rightarrow$ the simplest
0.14

🧠 **Tip: "Align the digits for addition and subtraction. Treat as an integer first for multiplication. After counting the decimal places, start from the right point."**

⚠️ **T2 most frequent error: $0.3 \times 0.4 = 1.2$ (forgetting to multiply decimals will get smaller and smaller!)**

Verification: 0.4 times of $0.3 < 0.3$, the answer must be < 0.3

KP1 Synchronization practice — more number of digits + decimal (question 1-10)

#	Question	Difficulty	Working Space
1	$3,456,789 + 2,543,211 = ?$		
2	$10,000,000 - 3,456,789 = ?$		
3	$0.45 + 0.8 = ?$ (Tip: Align the decimal points)		
4	$0.6 \times 0.7 = ?$		
5	$5,678,900 - 987,654 = ?$		
6	$0.25 \times 0.4 = ?$		
7	$23,456,789 + 87,654,321 = ?$		
8	$0.125 \times 0.8 = ?$		
9	$99,999,999 + 1 = ?$ (Test you: Where to carry?)		
10	$0.05 \times 0.02 = ?$ (How many zeros after the decimal point?)		

Knowledge Point 2: Score Calculation Trap Total Attack (T9) ● SSPA Required Exam

Five-step verification method for fraction calculation:

- ① **Look at the denominator** |||SEP|||: Same? Direct addition and subtraction. different? First common denominator Find LCM |||SEP|||: Use short division to find the least common multiple (not just find a common multiple!)
- ② **Common denominator** |||SEP|||: Numerator = original numerator $\times$ (LCM $\div$ original denominator) —must be expanded simultaneously
- ③ **Calculation**: The denominator remains unchanged, the numerators are added and subtracted **Reduction**
- ④ : **Find the HCF of the numerator and denominator $\rightarrow$ reduce to the simplest $\rightarrow$ convert improper fractions into mixed numbers**: The denominator remains unchanged, and the numerators are added and subtracted.
- ⑤ **reduce**: Find HCF of numerator and denominator $\rightarrow$ reduce to the simplest $\rightarrow$ convert improper fractions into mixed numbers

1

Looking for LCM

Short division least common multiple

2

common points

The numerator and denominator change simultaneously to LCM.

3

calculate

Adding and subtracting numerators without changing the denominator

4

reduce

Simplest false fraction → mixed fraction

Example KP2-1: Addition of different denominators

$$\frac{3}{8} + \frac{5}{12} = ?$$

✗ Common mistakes

$$\frac{8}{20}$$

Numerator + numerator, denominator + denominator: $3+5=8$, $8+12=20$ **✓ Correct solution**

$$\frac{19}{24}$$

$$\text{LCM}(8,12)=24 \rightarrow \frac{9}{24} + \frac{10}{24} = \frac{19}{24}$$

Example KP2-2: Subtraction with mixed numbers

$$2\frac{1}{3} - 1\frac{1}{2} = ?$$

✗ Common mistakes

$$2-1=1, \frac{1}{3} - \frac{1}{2} = \frac{1}{6} \rightarrow 1\frac{1}{6}$$

Subtract whole numbers and fractions separately, but $13-12$ should be $26-36=-16$ **✓ Correct solution**

$$\frac{5}{6}$$

Conversion to false fraction: $73-32 \rightarrow \text{LCM}(3,2)=6 \rightarrow 146-96=56$

Tip: "If you add points, turn them into false ones first, find LCM through points, and follow the multiplication of the numerator, the answer will definitely be the same."

KP2 synchronization practice — fractioncalculate (question 11-25)

#	Question	Difficulty	Working Space
11	$\frac{1}{4} + \frac{1}{8} = ?$		
12	$\frac{1}{3} + \frac{1}{9} = ?$		
13	$\frac{2}{5} + \frac{1}{3} = ?$		
14	$\frac{3}{4} - \frac{1}{6} = ?$		
15	$\frac{5}{6} - \frac{1}{4} = ?$		

16	$\frac{2}{3} + \frac{3}{5} = ?$		
17	$1\frac{1}{4} + 2\frac{2}{3} = ?$		
18	$3\frac{1}{2} - 1\frac{2}{3} = ?$		
19	$\frac{3}{8} + \frac{5}{12} = ?$		
20	$\frac{7}{10} - \frac{2}{5} = ?$		
21	$\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = ?$		
22	$2\frac{1}{5} + 1\frac{3}{10} = ?$		
23	$\frac{4}{9} + \frac{2}{3} = ?$		
24	$1\frac{2}{3} + 2\frac{3}{4} = ?$		
25	$\frac{8}{15} + \frac{7}{10} = ?$ (LCM(15,10)=?)		

Knowledge point 3: Comprehensive calculation strategy - verification = full score SSPA

Strategies for getting full marks on calculation questions (don't just complete the calculation, you need to verify!):

- ① **Estimation verification method** |||SEP|||: **Approximate mental arithmetic value (for example, $2.98 \times 4.1 \approx 3 \times 4 = 12$, the answer should be close to 12)** Inverse operation verification |||SEP|||: Use subtraction to test addition, and use division to test multiplication
- ② **Unit check**: Is the answer in the correct unit? Is the decimal point position reasonable?
- ③ **Mantissa check** |||SEP|||: **only look at the single digit/last digit, quick verification** Special value alert
- ④ **mantissa check**: Only look at single digits/last digit, quick verification
- ⑤ **Special value alert** : $\times 0.5 = \div 2$, $\div 0.5 = \times 2$, $\div 0.1 = \times 10$

Example question KP3-1: Estimation verification

Calculate $198 \times 52 = ?$ First use the estimate to determine which of the following answers is reasonable: A. 1,029 B. 10,296 C. 102,960

 **Choose without estimating**

Choose A or C

No habit of estimating = giving away points to a trap

 **Estimate verification**

B. 10,296

$198 \approx 200$, $200 \times 50 = 10,000 \rightarrow$ Only B is close

Example KP3-2: Mixing fractions and decimals

Calculate $\frac{1}{2} + 0.25 = ?$ (Convert to decimals or fractions)

✗ Error

Add directly: $\frac{1}{2} + 0.25 = 0.50.25 = ?$

Fractions and decimals cannot be added directly!

✓ Correct

$0.5 + 0.25 = 0.75 = \frac{3}{4}$

Convert to decimals: $12 = 0.5$, $0.5 + 0.25 = 0.75$

⚠ 35% of the points lost for the calculation questions in the first paper were due to "no verification" rather than "not knowing how to calculate". Get into the habit of estimating!

KP3 comprehensive calculate practice (question 26-35)

#	Question	Difficulty	Working Space
26	Estimate: $4,987 \times 31 \approx ?$ Which one is closest? A. 15,000 B. 155,000 C. 1,550,000		
27	$\frac{3}{4} + 0.5 = ?$ (Convert to decimal calculation)		
28	Verify with estimate: $12,345 + 87,655$ approximately = ?		
29	$0.75 + \frac{1}{4} = ?$ (Unified score calculation)		
30	$2.5 \times 4.2 = ?$ Use estimates first ($2.5 \times 4 = 10$), and then calculate arithmetic.		
31	Verify using the inverse operation: If $3,456 \div 12 = 288$, verify using multiplication.		
32	$\frac{2}{5} + 0.6 = ?$ (Unification: $0.4 + 0.6 = ?$)		
33	Estimate: $0.48 \times 9.7 \approx ?$ Which one is closest? A. 0.4656 B. 4.656 C. 46.56		
34	$1\frac{1}{4} + 0.75 = ?$ ($1.25 + 0.75 = ?$)		
35	Calculate $1 - \frac{1}{3} - 0.25 = ?$ (Add and subtract one after another, pay attention to the order)		

Sprint speed-up practice (question 36-45 · time limit 15 minutes)

#	Question	Difficulty	Working Space
36	$45,678 + 54,322 = ?$		
37	$0.8 \times 0.25 = ?$		
38	$\frac{3}{4} + \frac{1}{8} = ?$		
39	$1,000,000 - 456,789 = ?$		
40	$0.06 \times 0.5 = ?$		

41	$\frac{5}{6} - \frac{2}{9} = ?$		
42	$2,345 \times 67 = ?$ (Use estimates to verify the rationality of the answers)		
43	$\frac{1}{2} + \frac{2}{3} + \frac{3}{4} = ?$ (Three numbers can be divided into three numbers)		
44	$0.125 + \frac{3}{8} = ?$ ($0.125 = \frac{1}{8}$)		
45	$4\frac{1}{3} + 2\frac{5}{6} - 1\frac{1}{2} = ?$ (Mixed addition and subtraction - it is recommended to convert all improper fractions)		

III. The Lesson core common error summary

✓ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	trap	Correct Approach
1	Multi-digit addition and subtraction misaligned	T1	Units place to units place, tens place to tens place, hierarchical alignment
2	Carry/Borrow Missing Flag	T1	Carries are marked with a small number on the upper row, borrows are marked with a small dot on the borrowed number.
3	Decimal points are not aligned for decimal addition and subtraction	T2	The decimal points must be aligned, and any missing digits must be padded with 0
4	Forgetting the decimal point when multiplying decimals	T2	First multiply integers → count the total number of decimal places → click from right to left
5	$0.3 \times 0.4 = 1.2$	T2	The more you multiply a decimal, the smaller it gets: $0.3 \times 0.4 = 0.12$
6	Direct addition and subtraction of fractions with different denominators	T9	Must be divided first → LCM → simultaneous expansion of molecules
7	After the common denominator, only the denominator is changed and the numerator is forgotten.	T9	Numerator = original numerator $\times$ (LCM $\div$ original denominator)
8	The answer is not reduced	T9	Find HCF → reduce to the simplest → false → mixed fraction

9	Error in adding and subtracting mixed numbers	T9	Recommendation: Convert all false scores and then calculate
10	No verification habit	T1/T2/T9	Use estimates to determine the reasonableness of answers

Comprehensive method to submit calculation questions for score test **Must read**

Three principles of not doing:

- ① **Do not do it if it is not aligned |||SEP|||: Calculate multiple digits and decimals, never start counting if the digits are not aligned** Do not do it without common denominator |||SEP|||: Fractions with different denominators, LCM is not found, never start adding or subtracting
- ② **Do not hand in the paper without verification:** Calculate each question mentally once, if the answer is unreasonable, recalculate immediately
- ③ **Submission test strategy:** Calculation questions account for 40-50% of Paper 1, which is the easiest part to get full marks! Because there are standard answers, it is not an open question. Achieving "zero mistakes" in calculation questions = half of the success in the sub-test.

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 **Tips for calculating full marks: "Align the bits, divide first, then multiply the points, reduce the tail, and estimate the test."**